09 The Speciation Room

Divergence in isolation

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 37 min	**+ 32 min**	The Drift (04), Selection (05) and Migration (07) rooms. The Adaptation Room (08) supplies the traits.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#speciation`

An ancestral lake population carrying 1,013 loci of standing variation splits in two. Population A stays; population B moves to another lake or to a stream, and nothing passes between them again. After *G* generations the room counts fixed differences and the divergence of a mating signal, and gives a verdict. This guide asks what accumulates, where in the genome, and how much a verdict of this kind really says.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain why populations that never exchange genes accumulate fixed differences, even with no selection.
- Compare divergent selection (a stream) with parallel selection (another lake) at the loci each habitat can see.
- Show that mutations arrive in proportion to the number of loci, while selection changes which of them fix.
- Explain why a count of fixed differences is a model-defined outcome, not evidence of reproductive isolation.

The extension adds

- Explain the two regimes of divergence: sorting of standing variation, then new mutation.
- Use $E[H] = \theta/(2\theta + 1)$ and the ratio $4Ns$ to explain the numbers on the cards.
- Explain why a trait that is both under selection and used as a mating signal is a fast route to speciation.

SUGGESTED ROUTES

- **50-minute practical (BSc):** steps 1–4 and questions 1–3. Each run takes several seconds.
- **Capstone for Part III:** this room as the last practical of a course on population genetics, with question 3 as an essay prompt.
- **MSc:** steps 5–7 and questions 4–6 (about 35 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 A move to the stream

12 min

SET UP

The defaults: N = 200, N_2 = 50, G = 1000, B moves to **a stream**, θ = 0.16, s = 0.20, μ = 0.005, X = 154, colour as a mating signal **Yes**, Y = 75°.

PREDICT FIRST

Will A and B be declared two species after 1000 generations? If so, which barrier will be crossed first: the fixed-difference count or the colour divergence?

RUN

Press **Run Simulation**.

LOOK AT

the verdict, the two criteria charts (with their dashed red thresholds), and the three mean fish.

EXPLAIN

Which barrier went up first, and why might a trait that is under selection *and* used to choose mates be the faster one?

2 Another lake instead

10 min

SET UP

B moves to **another lake**. Everything else as before.

PREDICT FIRST

Will there be more or fewer fixed differences at the 13 morphological loci (the gold line) than in the stream? Will the genome-wide total (the black line) change much?

RUN

Run, then compare with your stream run.

LOOK AT

the gold, grey and black lines in the left criteria chart.

EXPLAIN

Why does the same habitat on both sides reduce divergence at the loci selection can see, but hardly move the total?

3 Where the new mutations went

10 min

SET UP
Either of your runs.

PREDICT FIRST
There are 13 morphological, 40 physiological and 960 neutral loci. In what proportions will *new* mutations land on them? Which class will turn the largest share of its mutations into substitutions?

RUN
Read the ledger *Where the new mutations went*, below the locus strip.

EXPLAIN
What does the ledger say about whether selection can make a useful mutation happen?

4 What the verdict does and does not mean

5 min

SET UP
Read the caveat beneath the populations.

PREDICT FIRST
If you ran two isolated populations in identical lakes for long enough, would the verdict eventually say *two species*?

EXPLAIN
Why is a count of fixed differences not proof of reproductive isolation? What would you need to measure in real animals?

EXTENSION · MSC

5 How repeatable is it?

12 min

SET UP
Defaults, stream; then another lake. Press **New Ancestral Population** between runs.

PREDICT FIRST
Out of five runs in each habitat, how many will end in a verdict of speciation?

RUN
Five runs in each habitat, recording the verdict and the morphological fixed differences each time.

EXPLAIN
Why do runs differ so much? Relate your answer to which variants each ancestor happened to carry.

6 A more variable ancestor

10 min

SET UP

Open *More parameters* and raise θ from 0.16 to **0.5**.

PREDICT FIRST

Calculate the ancestral heterozygosity $E[H] = \theta/(2\theta + 1)$ at both values. Will raising θ increase the fixed-difference count, the speed, or both?

RUN

Press **New Ancestral Population**, read the ancestral card, then run.

EXPLAIN

Why is standing variation a ceiling on how much divergence sorting alone can produce? What carries the count past that ceiling?

7 Small founding populations

10 min

SET UP

Back to $\theta = 0.16$. Set N_2 to **10**, then to **200**.

PREDICT FIRST

In which setting will B fix more new mutations of its own, and why? Which will exhaust its standing variation faster?

RUN

Run each, and compare the fixed-difference curves and the ledger.

EXPLAIN

Using the neutral fixation probability $1/(2N)$, explain why some of what separates A and B is drift in the smaller population rather than adaptation in either.

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Isolation is the *starting condition* of this room, not its result. Why does it have to be? Refer to the Migration Room.
2. BSc Two cichlid populations look almost identical but have rebuilt their physiology for different water. How does this room let that happen, and why would a photograph miss it?
3. BSc Write a short paragraph explaining to a non-biologist why “they have 150 genetic differences” does not by itself mean “they are different species”.
4. MSc Explain Dobzhansky–Muller incompatibilities, and why the room can only count them by proxy.
5. MSc The room has no linkage. How would a real genome, which sorts in blocks, change the pattern of divergence along the locus strip?
6. MSc Using the ratio $4Ns$, explain why a locus the habitat can see substitutes about 160 times as readily as one it cannot at the default settings.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** In the stream, speciation is declared in about four runs in five at the defaults. Colour is under selection there *and* is the mating signal, so the colour criterion is often the faster of the two barriers; the fixed-difference count needs many loci to sort and is slow.
- **Step 2.** The gold line reaches about 8 in the stream against about 5 between two lakes; the black total moves by only about nine of roughly 150, because 960 loci no habitat selects carry most of it.
- **Step 3.** New mutations land in proportion 13 : 40 : 960. The physiological loci collect around 28 fixed differences in the stream and 22 between lakes, five times more per locus than the neutral ones: selection changes fate, not supply.
- **Step 4.** Yes: drift alone accumulates fixed differences in isolated populations. Proof of speciation needs reproductive isolation itself — hybrids that fail, or individuals that will not mate.
- **Step 5.** Roughly four of five stream runs and one in three lake runs end in speciation; the spread comes from which few variants each ancestor carried.
- **Step 6.** E[H] rises from about 0.12 to 0.25. More standing variation means more loci available to sort, so more fixed differences, sooner. New mutation is what carries the count past that ceiling.
- **Step 7.** A small B fixes many more new mutations, mostly neutral ones, and exhausts its standing variation within a few dozen generations. Neutral fixation ($1/2N$) is far likelier in the small population.